

Sumedh Kothari

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EDUCATION

Georgia Institute of Technology

May 2028

B.S. in Chemical and Biomolecular Engineering

- **Coursework:** Computing for Engineers; Advanced Data Structures, Embedded Systems, and Networking; Multivariable Calculus; Linear Algebra; Intro Physics 1; Intro Physics 2
- **Honors:** Symposium of Rising Scholars Presenter, Eagle Scout, GPA: 3.8/4.0

EXPERIENCE

Georgia Tech Renewable Bioproducts Institute — Tong Laboratory | Research

Atlanta, GA

Research Intern

August 2025 – Present

- Applied Bayesian statistical modeling using Python and NumPy/Pandas libraries to optimize lignin valorization reaction conditions, improving process efficiency predictions
- Utilized NMR, FTIR, and DFT modeling data to characterize value-added chemicals, supporting process design intent for bio-based manufacturing
- Collaborated with research team to write manuscript for publication, demonstrating excellent written communication skills

Dressen Laboratory | Research

Los Altos, CA

Researcher

August 2024 - June 2025

- Designed and tested fluorinated Mg2dobpdc metal organic frameworks for vehicle exhaust CO2 capture, addressing environmental sustainability challenges
- Conducted adsorption efficiency, thermogravimetric analysis, NMR, and regeneration experiments to validate material performance
- Presented findings at SYNOPSIS 2025, showcasing strong oral communication and technical presentation abilities

Hydrogen Engineering Research Consortium | Research

Los Angeles, CA

Simulation Intern

August 2024 – February 2025

- Modeled species' concentrations in eutrophic freshwater ecosystem using MATLAB and dimensionless differential equations to analyze environmental systems
- Employed sustainability over sets calculation to determine equilibria and stability of environment, supporting environmental impact assessments
- Co-authored research paper on environmental systems modeling under UCLA faculty mentorship, demonstrating technical writing proficiency

ChemE Cube | Design

Atlanta, GA

Design Engineer

August 2025 - Present

- Designed high surface area Kelvin lattice structure for 1 cu. ft. Direct Air Capture Plant using AutoCAD, supporting process equipment selection
- Analyzed CO2 capture data with Python to determine optimal amine sorbent concentrations and solvent chemistry, generating process datasheets
- Prepared demo, technical documentation, and presentation for judging panel, ensuring process design intent was clearly communicated

AiApply | Development

Atlanta, GA

Developer

October 2025

- Developed full-stack AI-powered internship application platform using Python, Selenium WebDriver, HTML, and JavaScript to automate workflows
- Implemented Selenium-based automation to extract recruiter profiles and deliver personalized direct messages at scale, improving efficiency
- Engineered web scraping algorithms using Python to identify target companies and automate tailored email outreach

GreenGrade - Gemini API Developer Competition | Development

Los Angeles, CA

Project Lead

July 2024 - August 2025

- Directed 20-person team to architect and implement fuzzy logic-driven Sustainability Interval Index calculator with scalable data integration pipelines
- Developed and deployed GreenGrade (Flutter/C++), embedding SII computational model and automating sustainability analytics for 175 mining firms

- Utilized Gemini API to process and analyze sustainability data, demonstrating ability to work with advanced computational tools

Independent Research | Research
Researcher

Remote
January 2024 - June 2024

- Conducted 35-page literature review assessing chemical and economic feasibility of PFAS removal techniques for environmental remediation
- Collaborated with Yale's Dr. Julianne Rolf to validate research findings and methodology
- Presented at the Symposium of Rising Scholars, demonstrating technical communication and presentation skills

Sustainable Farming Crop Growth Predictor | Development
Developer

Remote
June 2022 – July 2022

- Implemented AI/ML models (MLP, linear regression, KNN, decision trees) in Python with NumPy/Pandas libraries to analyze crop growth data
- Optimized predictive analytics pipeline to identify high-performing farms by soil type, achieving 160% higher accuracy over random baseline
- Analyzed 30+ bacterial strain features to develop data-driven agricultural insights

PROJECTS

SKILLS

- **Technical:** Python, MATLAB, Excel, NumPy, Pandas, Selenium WebDriver, HTML, JavaScript, Flutter, C++, AutoCAD, NMR, FTIR, DFT Modeling, Bayesian Statistical Modeling, Process Flow Diagrams, Heat and Material Balances, Process Datasheets, Environmental Modeling, Data Analysis
- **Communication:** Technical Writing, Oral Communication, Multilingual (English, Hindi, Spanish), Team Leadership, Collaboration